Repeated Measures ANOVA

Confounding · Interaction · Assumption Testing · Model Fit

Using RStudio • ez • nlme • lme4 • emmeans

Dr Muhammad Adil Zainal Abidin

What is Repeated Measures ANOVA?

Analysing within-subject changes across multiple time points or conditions

Same participants measured ≥2 times → subjects are their own control → individual differences removed from error term

One-Way ANOVA

- Independent groups (between-subjects design)
- Each participant assigned to ONE group only
- Error = between-group + within-group variance
- Higher error term → lower statistical power
- e.g. Drug A vs Drug B vs Placebo (different patients)

Repeated Measures ANOVA

- Same subjects across all conditions or time points
- Within-subject design removes individual differences
- Error term = within-subject residual only
- Greater power, fewer subjects needed
- e.g. BP at Baseline, 1 month, 3 months (same patients)

Common designs: One-factor RM-ANOVA | Two-factor Mixed ANOVA (between + within) | Split-plot | MANOVA approach

Controlling for Confounders

RM-ANOVA automatically removes time-invariant confounders; between-subject confounders need explicit adjustment

Built-in Within-Subject Control

- ✓ Stable individual differences (genetics, baseline physiology)
- ✓ Fixed demographic factors (sex, ethnicity, blood type)
- ✓ Persistent lifestyle factors (socioeconomic status)
- ✓ Any time-invariant characteristic of the subject

How it works:

Subject-level variance is partitioned from the error term. Each person serves as their own control, eliminating the effect of any variable that does not change over time within a subject.

Between-Subject Confounders — Must Adjust

- → Add as between-subjects factor in ezANOVA
- → Include as fixed effect in LMM (nlme/lme4)
- → Baseline outcome as covariate (ANCOVA approach)

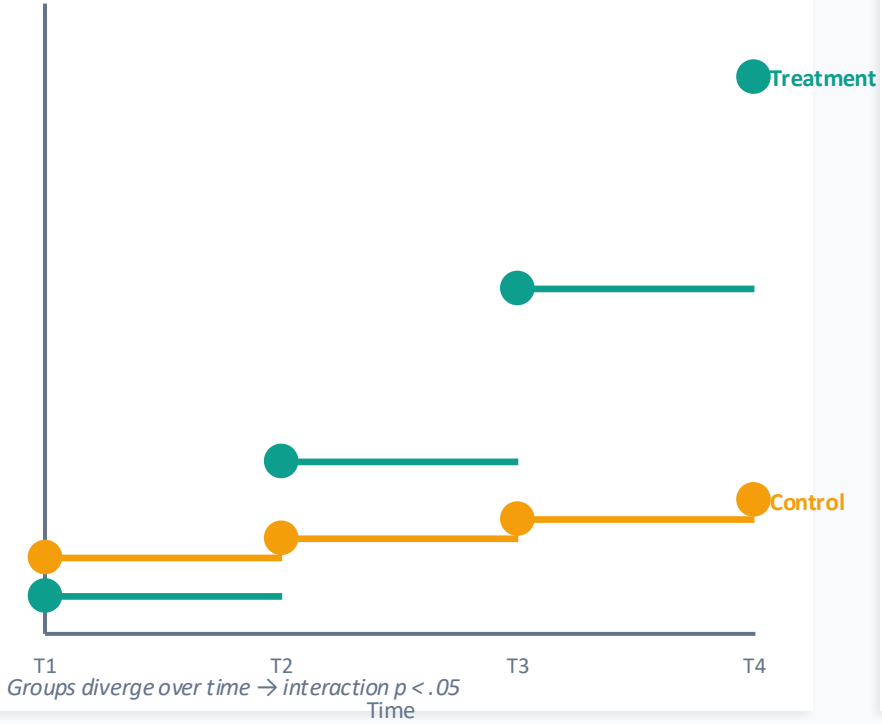
```
# Between-subject confounder as factor (ez)
library(ez)
ezANOVA(data=df, dv=score, wid=id,
  within  = time,
  between = group,    # confounder/treatment
  type = 3)

# ANCOVA style: baseline as covariate (nlme)
library(nlme)
lme(score ~ time * group + baseline,
    random = ~1|id, data = df)
```

Testing Interaction Effects

Time × Group interaction: do different groups change at different rates over time?

Significant Time × Group Interaction



Groups diverge over time → interaction p < .05

Types of Interaction

Time × Group

Main interaction: does treatment effect change over time? Central to RCTs and longitudinal studies.

Time × Covariate

Does a continuous variable (age, baseline) modify the time trajectory? Tests effect modification.

Three-way Interaction

e.g. Time × Group × Sex. Requires large n; interpret cautiously and report cautiously.

```
# Time × Group interaction (ez)
fit <- ezANOVA(data=df, dv=score, wid=id,
  within=time, between=group, type=3)
fit$ANOVA          # F, df, p, ges

# Post-hoc: simple effects per time point
library(emmeans)
lme_fit <- lme(score ~ time*group,
               random=~1|id, data=df)
emm <- emmeans(lme_fit, ~ group | time)
pairs(emm, adjust="bonferroni")
```

Comparing Analytical Methods

Four approaches — choose based on design complexity, missing data, and distributional assumptions

Classical RM-ANOVA

(ez / car)

✓ Pros

- Simple, widely understood output
- Built-in Mauchly sphericity test
- Automatic ε corrections (G-G, H-F)

✗ Cons

- Listwise deletion (no missing data)
- Sphericity may be violated
- No complex covariance structures

Use when:

Balanced, complete data; ≤4 time points

```
library(ez)
ezANOVA(data=df, dv=y,
  wid=id, within=time,
  between=group, type=3)
```

Linear Mixed Model

(nlme / lme4)

✓ Pros

- Handles missing data (REML)
- Flexible covariance structures
- Time-varying covariates allowed

✗ Cons

- More complex to specify
- Covariance choice critical
- No automatic post-hoc

Use when:

Unbalanced/missing data; complex designs

```
library(nlme)
lme(y ~ time*group,
  random=~1|id,
  corr=corAR1(),
  data=df, method="REML")
```

MANOVA Approach

(base R / car)

✓ Pros

- No sphericity assumption needed
- Controls FWER across time points
- Good for small number of time points

✗ Cons

- Requires n >> p (subjects > time pts)
- Low power with many time points
- Multivariate normality needed

Use when:

Sphericity violated; few time points

```
model <- lm(cbind(t1,t2,t3,t4)
            ~ group,
data=df_wide)
library(car)
Anova(model, type="III")
```

GEE

(geepack)

✓ Pros

- Population-average interpretation
- Robust to correlation mispecification
- Handles non-normal outcomes

✗ Cons

- Subject-specific inference not valid
- Less efficient than LMM if correct
- Requires large sample size

Use when:

Non-normal outcomes; population inference

```
library(geepack)
geeglm(y ~ time*group,
  id=id, data=df,
  corstr="exchangeable")
```

If sphericity violated: prefer LMM (nlme) with appropriate covariance structure over epsilon-corrected classical RM-ANOVA

Methods Comparison Table

Match your design requirements to the appropriate analytical method

Feature	RM-ANOVA (ez)	Linear Mixed Model	MANOVA	GEE
Missing data	✗ Listwise deletion	✓ Handles MAR	✗ Complete cases	✓ Handles MAR
Sphericity needed	✓ Tests & corrects	✗ Not required	✗ Not required	✗ Not required
Covariance structure	Compound symmetry	✓ AR1, UN, CS, Toep	Unstructured	Exch., AR1, Indep
Random effects	Subject as block only	✓ Slopes + intercepts	None	None
Interaction test	✓ Standard F-test	✓ Wald / LRT	✓ Pillai / Wilks	✓ Score test
Interpretation	Within-subj effect	Subject-specific	Multivariate	Population-average
Best for	Simple balanced RCTs	Longitudinal / LDA	Violated sphericity	Non-normal, large n

Testing Assumptions

Test all four before interpreting results — violations require specific remedies

1 Sphericity

What: Variances of differences between all pairs of time points must be equal

Test: Mauchly's test (W statistic). Auto-reported by ezANOVA.

Fix: p<.05: use Greenhouse-Geisser (ε<0.75) or Huynh-Feldt (ε≥0.75) corrected p-value

```
library(ez)
fit <- ezANOVA(data=df, dv=y, wid=id,
       within=time, type=3, return_aov=T)
```

2 Normality of Residuals

What: Within-subject residuals (or differences between time points) are normally distributed

Test: Shapiro-Wilk test (n<50), Q-Q plot, histogram of residuals

Fix: Log/square-root transform; use LMM or GEE for markedly non-normal outcomes

```
res <- residuals(lme_model)
shapiro.test(res)       # p > .05 = OK
qqnorm(res); qqline(res)
```

3 No Extreme Outliers

What: No observations that disproportionately influence the group means at any time point

Test: Boxplots per time point; standardised residuals |z|>3.29; Cook's distance

Fix: Investigate each outlier; run sensitivity analysis with and without; document decision

```
# Outlier detection per time point
boxplot(score ~ time, data=df)
# Cook's distance
```

4 Independence Between Subjects

What: Observations from different participants are independent (within-subject correlation is modelled)

Test: Verify by study design; ACF plot checks residual autocorrelation beyond modelled structure

Fix: If extra clustering exists, add nested random effects: random=~1|cluster/id

```
# Check residual autocorrelation
acf(residuals(lme_model))
# Bars within blue lines = OK
```

Complete RStudio Workflow

From data preparation to post-hoc tests — step by step

```
# ── 1. RESHAPE TO LONG FORMAT ──────────────────────────────────
library(tidyverse)
df_long <- df_wide %>%
  pivot_longer(cols=c(t0,t1,t2,t3),
               names_to="time", values_to="score") %>%
  mutate(time=factor(time), id=factor(id), group=factor(group))
# ── 2. ASSUMPTION CHECKS ───────────────────────────────────────
df_long %>% group_by(time) %>%
  summarise(sw_p = shapiro.test(score)$p.value)  # normality per tp
boxplot(score ~ time, data=df_long)               # outliers
# ── 3. FIT RM-ANOVA (ez) ───────────────────────────────────────
library(ez)
fit <- ezANOVA(data=df_long, dv=score, wid=id,
               within=time, between=group,
               type=3, return_aov=TRUE)
fit$ANOVA                              # F, df, p, ges
fit$"Mauchly's Test for Sphericity"    # W, p
fit$"Sphericity Corrections"           # GGe, HFe corrected p
# ── 4. FIT LINEAR MIXED MODEL (nlme) ───────────────────────────
library(nlme)
lme_fit <- lme(score ~ time * group,
               random = ~1|id,
               correlation = corAR1(form=~1|id),
               data = df_long, method="REML")
summary(lme_fit)      # fixed effects + variance components
intervals(lme_fit)    # 95% CI
# ── 5. POST-HOC COMPARISONS ────────────────────────────────────
library(emmeans)
emm <- emmeans(lme_fit, ~ time | group)
pairs(emm, adjust="bonferroni")       # pairwise tests
contrast(emm, method="pairwise")
```

Key Output Reference

F, df, p (ANOVA)
Report with effect size ges; note if GG/HF correction applied

Mauchly W & p
p<.05 = sphericity violated; state which correction used

Fixed effects β
Coefficient, SE, t, p and 95% CI for each term in LMM

Random effects σ^2
Between-subject variance; compute ICC = σ^2_subj / σ^2_total

emmeans pairs()
Bonferroni-adjusted pairwise p-values with Cohen's d

AIC / BIC
Compare covariance structures; ΔAIC > 2 = meaningfully better fit

Data must be in LONG format for all packages

Assessing Model Fit

Use multiple criteria together — no single metric tells the whole story

η^2/ges

Effect Size (ges)

Generalised eta-squared (ges) from ezANOVA: Small=0.01, Medium=0.06, Large=0.14. Prefer ges over partial η^2 in RM designs.

```
fit$ANOVA$ges
```

R^2m/c

Marginal & Conditional R^2

R^2marginal = variance from fixed effects only. R^2conditional = fixed + random. Use MuMIn package for LMM R^2 values.

```
library(MuMIn)
```

AIC

AIC / BIC (LMM)

Compare covariance structures and random effects. Lower = better. ΔAIC > 2 indicates a meaningfully better fitting model.

```
AIC(m_cs, m_ar1, m_un)
```

diag

Residual Diagnostics

Residuals vs fitted (homoscedasticity), Q-Q plot (normality), ACF plot (autocorrelation). Run plot(lme_fit) for all-in-one check.

```
plot(lme_fit)          # resid vs fitted
```

ICC

Intraclass Correlation

ICC = σ^2_subject / (σ^2_subject + σ^2_residual). High ICC (>0.5) confirms RM design is more efficient than independent groups ANOVA.

```
library(performance)
```

Reporting Checklist

What to include in Methods and Results sections — based on APA and CONSORT guidelines

Design & Software

- State design: one-factor RM, mixed ANOVA, split-plot
- R version and package versions (ez, nlme, emmeans)
- State method: classical RM-ANOVA or LMM; justify choice
- Covariance structure chosen and how (AIC comparison)

Assumption Checks

- Mauchly's W statistic and p-value
- If violated: state ε value and correction applied (G-G / H-F)
- Shapiro-Wilk W and p for residual normality per group
- Confirm no extreme outliers or justify their retention

Main Results

- F(df1, df2), p-value for each main effect and interaction
- Effect size: ges for RM-ANOVA; R^2m and R^2c for LMM
- ICC value with 95% CI (justifies random effects structure)
- Profile plot: estimated marginal means ± SE across time

Post-hoc & Contrasts

- Correction method: Bonferroni, Holm, or FDR — state which
- Pairwise comparisons: emmeans estimate, 95% CI, adjusted p
- Cohen's d for each pairwise comparison (effect size)
- Simple effects: group differences at each time point

Pipeline: pivot_longer → ezANOVA (sphericity) → lme if needed → emmeans (post-hoc) → r.squaredGLMM + icc (model fit)